

# Bill of Materials

Project Title: VISIR

Board Name: VISB

Board P/N: 20210001001

# Ionosens

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| Part Number         | Description                                                                                | Designator                                                                 | Manufacturer     | Manufacturer P/N  | Quantity |
|---------------------|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|------------------|-------------------|----------|
| C01-0402-01-00001-1 | Cap Ceramic 100nF 50V X7R 5% SMD 0402 125°C                                                | C1, C2, C3, C4, C5, C6, C7, C8, C9, C10, C11, C12, C13, C14, C16, C17, C20 | YAGEO            | CC0402JRX7R7BB104 | 17       |
| C01-0402-01-00004-1 | Cap Ceramic 1uF 16V XR7 5% SMD 0603 125°C                                                  | C15                                                                        | YAGEO            | CC0603JRX7R7BB105 | 1        |
| C01-0402-01-00008-1 | Cap Ceramic 20pF 50V C0G 5% SMD 0402 125°C                                                 | C18, C19, C22, C23                                                         | YAGEO            | CC0402JRNPO9BN200 | 4        |
| C01-0402-01-00002-1 | Multilayer Ceramic Capacitors (High dielectric type), 16V, 100 uF, ± 20 %, 1210, X5R, RoHS | C21                                                                        | Taiyo Yuden      | EMK325ABJ107MM-P  | 1        |
| C01-0402-01-00005-1 | 1210 100 V 1 uF X7R ±10% Tolerance Multilayer Ceramic Chip Capacitor                       | C24                                                                        | YAGEO            | CC1210KKX7R0BB105 | 1        |
| C01-0402-01-00003-1 | Cap Ceramic 27nF 50V X7R 5% SMD 0805 125°C                                                 | C25                                                                        | YAGEO            | CC0805JRX7R9BB273 | 1        |
| CMP-004-00031-2     | DIODE SCHOTTKY 30V 1A SOD123 / Rectifier Diode Schottky 30V 1A 2-Pin SOD-123               | D1                                                                         | ON Semiconductor | MBR130T1G         | 1        |

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| CMP-004-00033-1     | Schottky Rectifier, 100V, 20A,<br>TO-252AE Rohs Compliant:<br>Yes                                                     | D2, D4, D6                                                                                                                              | Vishay          | V20PW10-M3/I                    | 3  |
| CMP-004-00032-1     | Diode Tvs Vwm 36V 1.5Kw<br>Bidir Smc Rohs Compliant                                                                   | D3, D5, D9                                                                                                                              | Bourns          | SMCJ36CA                        | 3  |
| D01-0402-01-00001-1 | Led red 0402                                                                                                          | D7, D10, D12                                                                                                                            | ROHM            | SML-P12UTT86R                   | 3  |
| CMP-004-00035-1     | Diode Tvs Vwm 7V 1.5Kw Bidir<br>Smc Rohs Compliant                                                                    | D8                                                                                                                                      | Bourns          | SMCJ7.0CA                       | 1  |
| D01-0402-01-00000-1 | Led green 0402                                                                                                        | D11, D13, D14, D15, D16,<br>D17, D18, D19, D20, D21,<br>D22, D23, D24, D25, D26,<br>D27, D28, D29, D30, D31,<br>D32, D33, D34, D35, D36 | ROHM            | SML-P13PTT86R                   | 25 |
| CMP-005-00013-1     | PTC Resettable Fuse<br>0.5A(hold) 1A(trip) 15V 100A<br>0.8W 0.15s 0.15Ohm SMD<br>Solder Pad 1812                      | F1                                                                                                                                      | Bourns          | MF-MSMF050-2                    | 1  |
| CMP-005-00014-1     | PTC Resettable Fuse<br>1.1A(hold) 2.2A(trip) 33VDC<br>100A 1.5W 0.5s 0.09Ohm<br>SMD Solder Pad 2920 T/R<br>Automotive | F2, F3                                                                                                                                  | Belfuse         | OZCF0110FF2C                    | 2  |
| CMP-005-00015-1     | Low Ohmic PTC Resettable<br>Fuse 5.5A 12V 1210 T/R                                                                    | F4                                                                                                                                      | Bourns          | MF-USML550/12-2                 | 1  |
| J01-001-01-00001-2  | Conn Socket and Terminal<br>Strip HDR/SKT 100 POS<br>0.5mm Solder ST SMD T/R                                          | J1, J2                                                                                                                                  | Samtec          | LSHM-150-06.0-L-DV-A-N-K-<br>TR | 2  |
| J01-001-01-00006-1  | Pluggable Terminal Blocks 6<br>Pos 5mm pitch Thro ugh Hole<br>Header                                                  | J3                                                                                                                                      | Phoenix Contact | 1776731                         | 1  |
| J01-001-01-00003-2  | 6 Vias Pin Header 2.54mm,<br>Right Angle                                                                              | J4, J11                                                                                                                                 | Samtec          | TSW-103-08-F-D-RA               | 2  |

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| J01-001-01-00007-2  | Pluggable Terminal Blocks 4 Pos 5mm pitch Thro ugh Hole Header                           | J5                                                                                    | Phoenix Contact    | 1776715            | 1  |
| J01-001-01-00005-2  | 6 Vias Pin Header 2.54mm, Right Angle, 1 Row                                             | J6, J12                                                                               | Samtec             | HTSW-106-08-F-S-RA | 2  |
| J01-001-01-00008-1  | Pluggable Terminal Blocks 4 Pos 5.08mm pitch T hrough Hole Header                        | J7                                                                                    | Phoenix Contact    | 1776524            | 1  |
| J01-001-01-00004-1  | Conn USB 2.0 Type B RCP 4 POS 2.5mm Solder RA Thru-Hole 4 Terminal 1 Port Box/Tray       | J8                                                                                    | TE Conectivity     | 292304-1           | 1  |
| J01-001-01-00009-2  | Conn SMA RCP 0Hz to 18GHz 50Ohm Solder RA Thru-Hole Gold                                 | J9, J10                                                                               | Samtec             | SMA-J-P-H-RA-TH1   | 2  |
| K01-0008-01-00002-6 | Reed Relay 5VDC 350Ohm 0.5A SPST-NO THT                                                  | K1, K16                                                                               | COTO               | 9002-05-00         | 2  |
| K01-0008-01-00008-1 | Reed Relay, Mini SIP, 5 VDC; 1 Form A, 15W, 200 VDC, 1.25A                               | K2, K3, K4, K5, K6, K7, K8, K9, K10, K11, K12, K13, K14, K15, K17, K18, K19, K20, K21 | Comus              | 3570.1419.051      | 19 |
| R01-0805-01-00691-1 | Ferrite Beads Multi-Layer High Current 1KOhm 25% 100MHz 1.6A 0.12Ohm DCR 0805            | L1, L2                                                                                | Murata Electronics | BLM21SP102SN1D     | 2  |
| CMP-00028-00007-2   | hole 5mm pad drill 2.2                                                                   | MH1, MH2, MH3, MH4                                                                    | Ionosens           | H5D22              | 4  |
| CMP-013-00034-1     | 100-V, N channel NexFET™ power MOSFET, single D2PAK, 3.4 mOhm 3-DDPAK/TO-263 - 55 to 175 | Q1                                                                                    | Texas Instrument   | CSD19535KTT        | 1  |
| CMP-013-00035-1     | Transistor MOSFET P-Channel 30V 7.5A 3W (Ta) Surface Mount SOT-223-4                     | Q2                                                                                    | ON Semiconductor   | NDT456P            | 1  |

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| CMP-013-00021-4     | N-Channel Logic Level Enhancement Mode Field Effect Transistor, 220 mA, 50 V, -55 to 150 degC, 3-Pin SOT-23, RoHS, Tape and Reel | Q3                                                                                                                                         | Fairchild | BSS138           | 1  |
| R01-0402-01-00169-2 | RES SMD 10K OHM 1% 1/16W 0402                                                                                                    | R1, R5, R7, R9                                                                                                                             | Yageo     | AC0402FR-0710KL  | 4  |
| R01-0402-01-00227-3 | RES SMD 22 OHM 1% 1/16W 0402                                                                                                     | R2, R3                                                                                                                                     | Yageo     | AC0402FR-0722RL  | 2  |
| R01-0402-01-00105-2 | RES SMD 4.7K OHM 1% 1/16W 0402                                                                                                   | R4, R6                                                                                                                                     | Yageo     | AC0402FR-074K7L  | 2  |
| R01-0402-01-00510-2 | RES SMD 499K OHM 1% 1/16W 0402                                                                                                   | R8                                                                                                                                         | Yageo     | AC0402FR-07499KL | 1  |
| R01-0402-01-00418-2 | RES SMD 196K OHM 1% 1/16W 0402                                                                                                   | R10                                                                                                                                        | Yageo     | AC0402FR-07196KL | 1  |
| R01-0402-01-00437-2 | RES SMD 240 OHM 1% 1/16W 0402                                                                                                    | R11, R13, R16, R20, R21, R22, R23, R24, R25, R26, R27, R28, R29, R30, R31, R32, R33, R34, R35, R36, R37, R38, R39, R40, R41, R42, R43, R44 | Yageo     | AC0402FR-07240RL | 28 |
| R01-0402-01-00196-4 | RES SMD 16.2K OHM 1% 1/16W 0402                                                                                                  | R12                                                                                                                                        | Yageo     | AC0402FR-0716K2L | 1  |
| R01-0402-01-00066-2 | RES SMD 2.94K OHM 1% 1/16W 0402                                                                                                  | R14                                                                                                                                        | Yageo     | AC0402FR-072K94L | 1  |
| R01-0402-01-00471-2 | RES SMD 332K OHM 1% 1/16W 0402                                                                                                   | R15                                                                                                                                        | Yageo     | AC0402FR-07332KL | 1  |
| R01-0402-01-00223-2 | RES SMD 22.6K OHM 1% 1/16W 0402                                                                                                  | R17                                                                                                                                        | Yageo     | AC0402FR-0722K6L | 1  |
| R01-0402-01-00360-2 | RES SMD 100K OHM 1% 1/16W 0402                                                                                                   | R18                                                                                                                                        | Yageo     | AC0402FR-07100KL | 1  |

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| R01-0603-01-00000-4   | RES SMD 0 OHM 1% 1/10W 0603                                                                            | R19        | Yageo             | AC0603FR-070RL  | 1   |
| CMP-0005-0001-00001-1 | 50-V, 7-ch darlington transistor array, -40 to 105C 16-TSSOP -40 to 105                                | U1, U4, U5 | Texas Instruments | ULN2003AIPWR    | 3   |
| U01-0007-01-00002-1   | 8-bit AVR MCU, 32 KB, 2560 Bytes, 16 MHz, 2.7 to 5.5V, TQFP-44, RoHS                                   | U2         | Microchip/Atmel   | ATMEGA32U4-AU   | 1   |
| U01-0007-01-00000-1   | AVR ATmega Microcontroller 8-bit 16MHz 256KB FLASH TQFP100                                             | U3         | Microchip         | ATMEGA2560-16AU | 1   |
| CMP-00023-00003-1     | DS1682 Series 5.5 V I2C 2-Wire Serial Surface Mount Real Time Clock - SOIC-8                           | U6         | Maxim             | DS1682S+        | 1   |
| CMP-001-00084-4       | 4.5-V to 60-V, 31mΩ, 0.6-6A eFuse with output power limiting and surge protection 20-HTSSOP -40 to 125 | U7         | Texas Instruments | TPS26631PWPR    | 1   |
| Y01-0001-01-00000-11  | 16MHz ±10ppm 16pF 80Ω SMD-2520_4P                                                                      | Y1, Y2     | SEIKO-EPSON       | Q24FA20H0024700 | 2   |
| Z01-0603-01-00000-1   | ESD Suppressor Multilayer Varistor 5V Automotive 2-Pin Case 0603                                       | Z1, Z2     | Bourns            | CG0603MLC-05E   | 2   |
|                       |                                                                                                        |            |                   |                 | 162 |

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| Approved | Notes |
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